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Course ID-Section: IT228-G5

Laboratory Activity 6

Laboratory Activity 6A

Configuring Router-on-a-Stick Inter-VLAN Routing

Part 1: Gather Information with Show Commands

Step 2: What process failed and why? The ping process failed because PC1 and PC3 are on different VLANs (and different IP networks). Without a router or Layer 3 device to perform inter-VLAN routing, hosts on separate VLANs cannot communicate with each other.

Part 2: Add VLANs to a Switch

Step 3: From **PC1**, ping **PC3**. The pings should still fail. Why were the pings unsuccessful? he pings were unsuccessful because the switch port connected to the router (G0/1) has not yet been configured as a trunk port. Without trunking, the router's subinterfaces cannot carry traffic for multiple VLANs, so inter-VLAN routing cannot function even though the subinterfaces are already configured on R1.

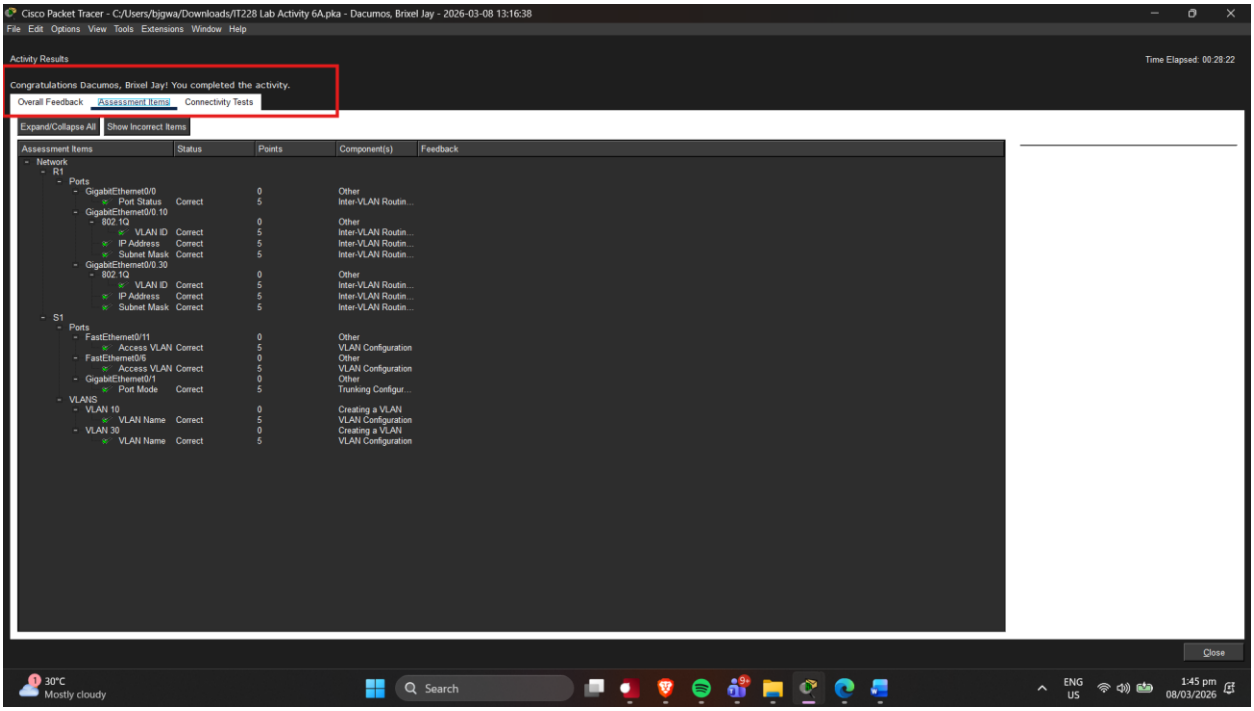
Part 4: Test Connectivity With Inter-VLAN Routing

Step 2:

a. On **S1**, issue the **show vlan** command. What VLAN is G0/1 assigned to? G0/1 is assigned to VLAN 1 (the default VLAN)

c. How can you determine that the interface is a trunk port using the **show vlan** command? You can determine it is a trunk port because the interface (G0/1) no longer appears in the show vlan output. Trunk ports are excluded from the VLAN port listing since they carry traffic for all VLANs rather than belonging to just one.

Screenshot of Assessment Items (with the congratulatory message that includes your name, and score on a red box):



Laboratory Activity 6B

Troubleshooting Inter-VLAN Routing

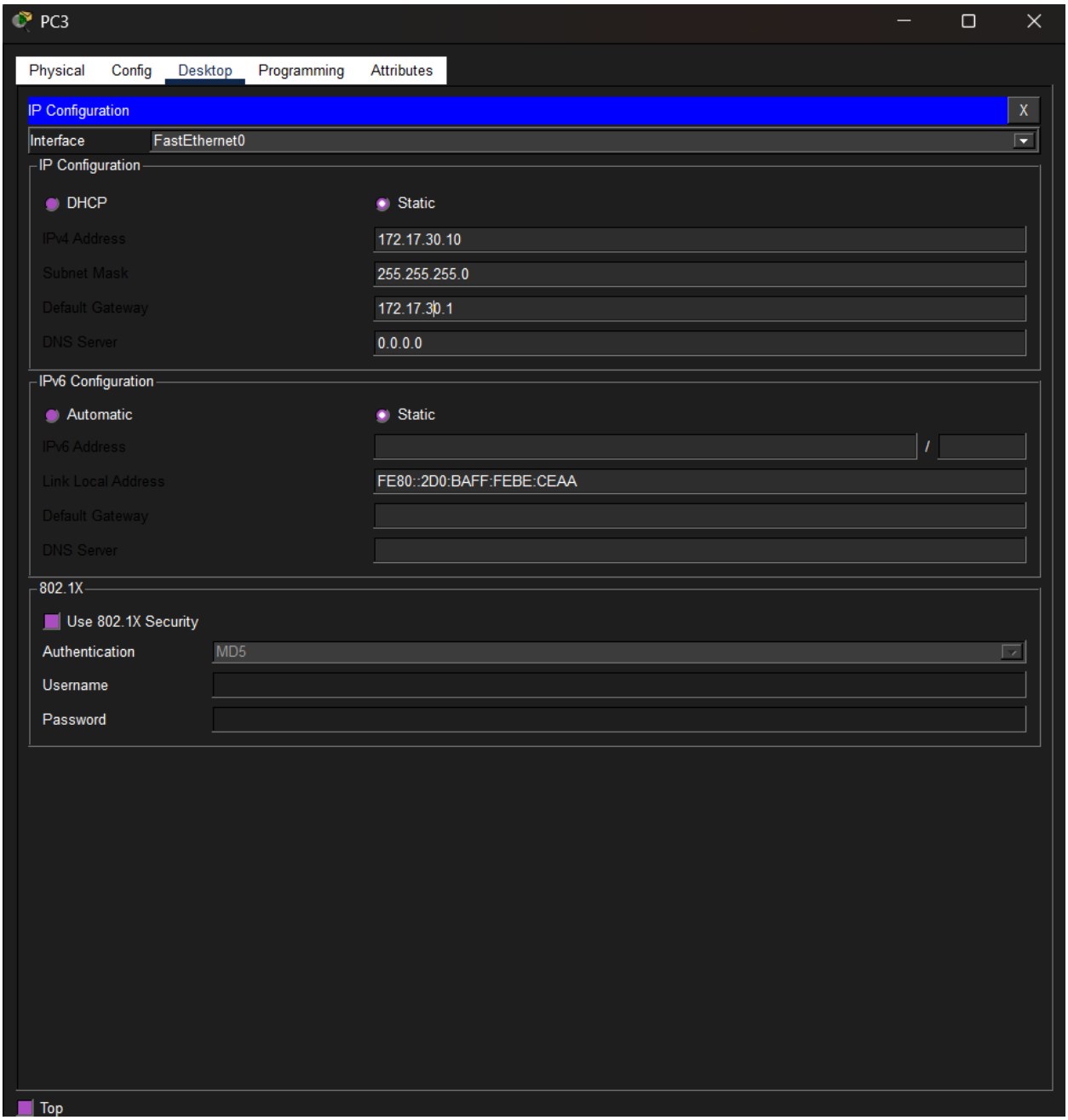
<u>Documentation Table</u>

<u>Problems</u>	**<u>Solutions</u>**
PC3 was configured with the wrong default gateway (172.17.10.1) which belongs to VLAN 10, not VLAN 30	Update PC3's default gateway to 172.17.30.1 to match its correct VLAN 30 network
The G0/1.10 subinterface on R1 was not active and could not pass traffic	Bring up the physical G0/1 interface using no shutdown so the subinterfaces become operational
The G0/1 port on S1 was set as an access port instead of a trunk port	Reconfigure S1's G0/1 port to trunk mode so it can carry traffic from multiple VLANs simultaneously
The VLAN encapsulation on R1's subinterfaces were swapped or	Delete the incorrect subinterface configurations and reapply the proper

incorrectly assigned, not matching the addressing table	`encapsulation dot1Q` commands along with the correct IP addresses for each subinterface

Screenshots of Troubleshooted Problems and Solutions:

Explain the significance of adding the steps that you took a screenshot with. Failure to attach the steps and explaining each step will be subject to deduction of your final score for this activity.



The significance of this Step is to match the default gateway of PC3 to the according default gateway match to the table.

R1

Physical Config CLI Attributes

```
1    default                          active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
R1#show inter
R1#show interfaces trunk


R1#config t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#no int g0/1.10
R1(config)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/1.10, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1.10, changed state to down

R1(config)#no int g0/1.30
R1(config)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/1.30, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1.30, changed state to down

R1(config)#int g0.1.10
                      ^
% Invalid input detected at '^' marker.

R1(config)#int g0/1.10
R1(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/1.10, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1.10, changed state to up

R1(config-subif)#encap
R1(config-subif)#encapsulation dot1Q 10
R1(config-subif)#ip add
R1(config-subif)#ip address 172.17.10.1 255.255.255.0
R1(config-subif)#ex
R1(config)#int g0/1.30
R1(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/1.30, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1.30, changed state to up

R1(config-subif)#ecap
R1(config-subif)#encap
R1(config-subif)#encapsulation dot1Q 30
R1(config-subif)#ip add
R1(config-subif)#ip address 172.17.30.1 255.255.255.0
R1(config-subif)#ex
R1(config)#int g0/1
R1(config-if)#no sh
R1(config-if)#ex
R1(config)#
```

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Enabling the subinterface is critical because it activates the **logical pathway** required for

VLAN 10 traffic to reach the router for inter-VLAN routing.

S1

Physical Config CLI Attributes

```
%LINK-5-CHANGED: Interface FastEthernet0/11, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/11, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/6, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/6, changed state to up

%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up


S1>en
S1#config t
Enter configuration commands, one per line.  End with CNTL/Z.
S1(config)#int g0/1
S1(config-if)#switchport mode trunk

S1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

S1(config-if)#ex
S1(config)#ex
S1#
%SYS-5-CONFIG_I: Configured from console by console

S1#show vlan

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                                Fa0/5, Fa0/7, Fa0/8, Fa0/9
                                                Fa0/10, Fa0/12, Fa0/13, Fa0/14
                                                Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                                Fa0/19, Fa0/20, Fa0/21, Fa0/22
                                                Fa0/23, Fa0/24, Gig0/2
10   Faculty/Staff                    active    Fa0/11
30   Guest(Default)                   active    Fa0/6
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active

VLAN Type  SAID       MTU   Parent RingNo BridgeNo Stp  BrdgMode Trans1 Trans2
---- ----- ---------- ----- ------ ------ -------- ---- -------- ------ ------
1    enet  100001     1500  -      -      -        -    -        0      0
10   enet  100010     1500  -      -      -        -    -        0      0
30   enet  100030     1500  -      -      -        -    -        0      0
1002 fddi  101002     1500  -      -      -        -    -        0      0
 --More--
```

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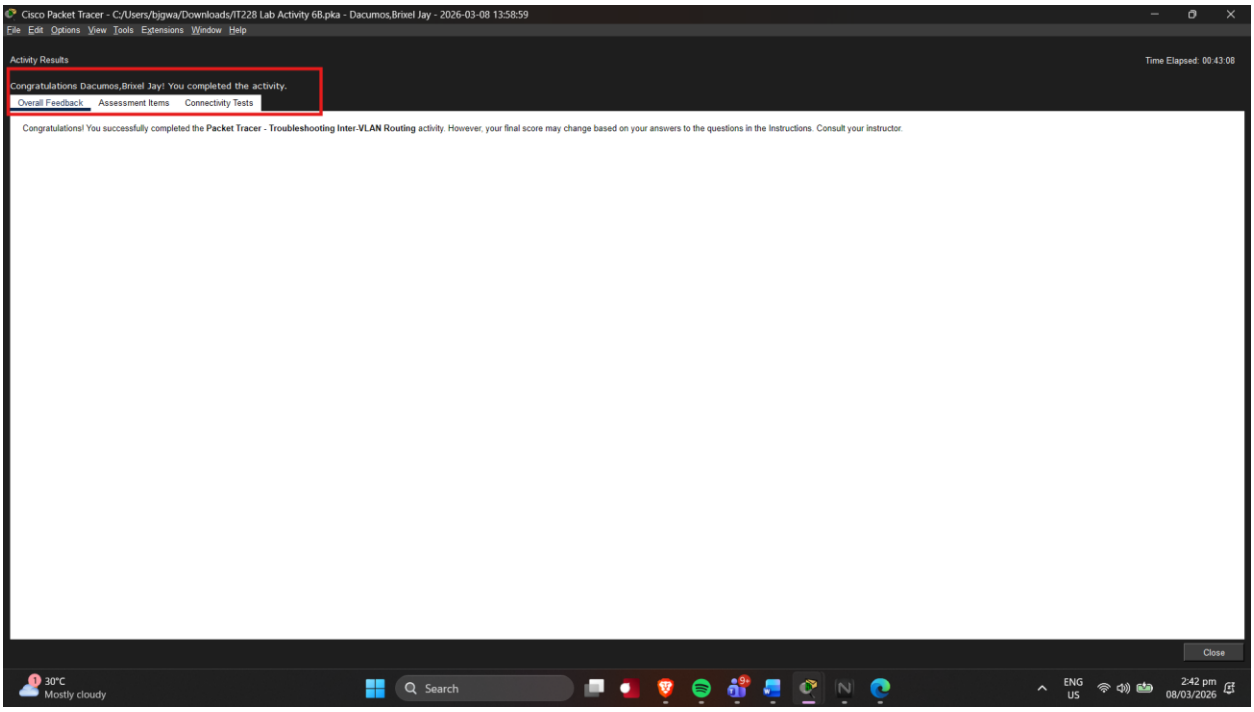
Screenshot of Check Results(with the congratulatory message that includes your name on a red box):


Cisco Packet Tracer - C:/Users/bjgwa/Downloads/IT228 Lab Activity 6B.pka - Dacumos,Brixel Jay - 2026-03-08 13:58:59
File Edit Options View Tools Extensions Window Help
Activity Results
Time Elapsed: 00:43:08
Congratulations Dacumos,Brixel Jay! You completed the activity.
Overall Feedback
Assessment Items
Connectivity Tests
Congratulations! You successfully completed the Packet Tracer - Troubleshooting Inter-VLAN Routing activity. However, your final score may change based on your answers to the questions in the Instructions. Consult your instructor.
Close
30°C
Mostly cloudy
Search
ENG
US
2:42 pm
08/03/2026